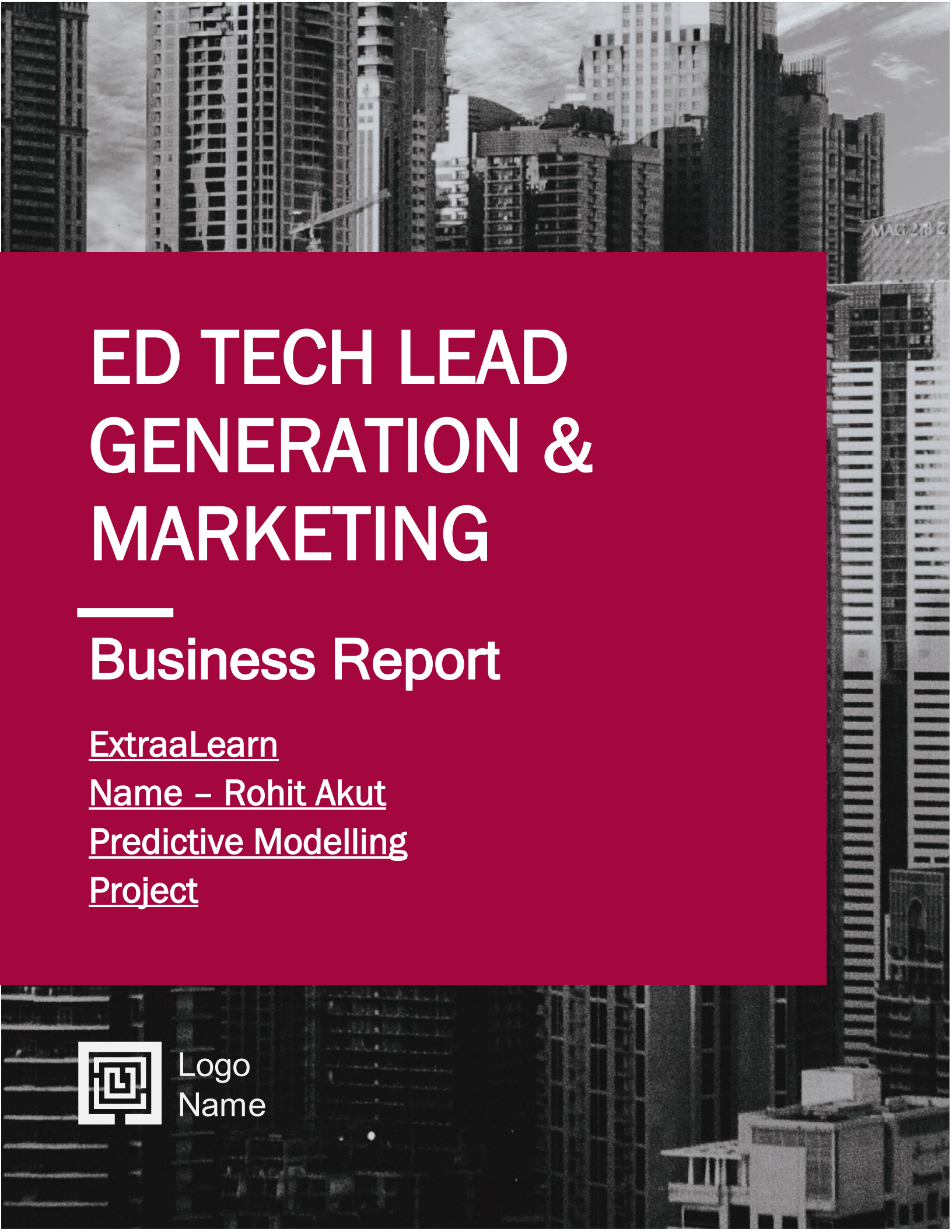


ED TECH LEAD GENERATION & MARKETING

Business Report

ExtraaLearn

Name – Rohit Akut

Predictive Modelling

Project

Logo Name

Table of contents

1. Summary
2. Introduction
3. Data Description
4. Sample Dataset
5. Exploratory Data Analysis
6.

Summary

ExtraaLearn is an initial stage startup that offers programs on cutting-edge technologies to students and professionals to help them upskill/reskill. With many leads being generated on a regular basis, one of the issues faced by ExtraaLearn is to identify which of the leads are more likely to convert so that they can allocate resources accordingly. Based on the analysis of different means of lead generation we can identify which ones are the most important and crucial in building a majority clientele.

Introduction

The purpose of this exercise is to explore the dataset. Do the exploratory data analysis. Explore the dataset using model generation, LDA, Decision tree and regression analysis and other parameters. The data contains the different attributes of leads and their interaction details with ExtraaLearn. The data lists 4612 individual leads from various modes of customer contact which include both online and offline modes.

Problem Statement

ExtraaLearn is an initial stage startup that offers programs on cutting-edge technologies to students and professionals to help them upskill/reskill. With a large number of leads being generated on a regular basis, one of the issues faced by ExtraaLearn is to identify which of the leads are more likely to convert so that they can allocate resources accordingly. You, as a data scientist at ExtraaLearn, have been provided the leads data to:

- Analyze and build an ML model to help identify which leads are more likely to convert to paid customers.

Data Description

- ID: ID of the lead
- age: Age of the lead
- current_occupation: Current occupation of the lead. Values include 'Professional','Unemployed',and 'Student'
- first_interaction: How did the lead first interacted with ExtraaLearn. Values include 'Website', 'Mobile App'
- profile_completed: What percentage of profile has been filled by the lead on the website/mobile app. Values include Low - (0-50%), Medium - (50-75%), High (75-100%)
- website_visits: How many times has a lead visited the website
- time_spent_on_website: Total time spent on the website in seconds
- page_views_per_visit: Average number of pages on the website viewed during the visits.
- last_activity: Last interaction between the lead and ExtraaLearn.
- Email Activity: Seeking for details about the program through email, Representative shared information with the lead like brochure of program, etc
- Phone Activity: Had a Phone Conversation with the representative, Had conversation over SMS with the representative, etc
- Website Activity: Interacted on live chat with a representative, Updated profile on the website, etc
- print_media_type1: Flag indicating whether the lead had seen the ad of ExtraaLearn in the Newspaper.
- print_media_type2: Flag indicating whether the lead had seen the ad of ExtraaLearn in the Magazine.
- digital_media: Flag indicating whether the lead had seen the ad of ExtraaLearn on the digital platforms.
- educational_channels: Flag indicating whether the lead had heard about ExtraaLearn in the education channels like online forums, discussion threads, educational websites, etc.
- referral: Flag indicating whether the lead had heard about ExtraaLearn through reference.
- status: Flag indicating whether the lead was converted to a paid customer or not.

Sample Dataset

	ID	age	current_occupation	first_interaction	profile_completed	website_visits	time_spent_on_website	page_views_per_visit	last_activity	print_media_t
0	EXT001	57	Unemployed	Website	High	7	1639	1.86100	Website Activity	
1	EXT002	56	Professional	Mobile App	Medium	2	83	0.32000	Website Activity	
2	EXT003	52	Professional	Website	Medium	3	330	0.07400	Website Activity	
3	EXT004	53	Unemployed	Website	High	4	464	2.05700	Website Activity	
4	EXT005	23	Student	Website	High	4	600	16.91400	Email Activity	

Fig 1 Sample dataset

The dataset has 15 variables with modes of client interaction and other attributes about the contact with respect to individual hits and the mode of contact.

Exploratory Data Analysis

Types of variables and Data types in the data frame:

```
Data columns (total 15 columns):
 #   Column               Non-Null Count  Dtype
---  ------               --------------  -----
 0   ID                   4612 non-null   object
 1   age                  4612 non-null   int64
 2   current_occupation   4612 non-null   object
 3   first_interaction    4612 non-null   object
 4   profile_completed    4612 non-null   object
 5   website_visits       4612 non-null   int64
 6   time_spent_on_website  4612 non-null   int64
 7   page_views_per_visit  4612 non-null   float64
 8   last_activity        4612 non-null   object
 9   print_media_type1    4612 non-null   object
 10  print_media_type2    4612 non-null   object
 11  digital_media        4612 non-null   object
 12  educational_channels  4612 non-null   object
 13  referral             4612 non-null   object
 14  status               4612 non-null   int64
```

There are 15 columns and 4612 rows in the dataset, out of 15 columns 10 are of Object data type while 4 are integer and 1 is of float data type.

Statistical Summary of Data

	age	website_visits	time_spent_on_website	page_views_per_visit	status
count	4612.00000	4612.00000	4612.00000	4612.00000	4612.00000
mean	46.20121	3.56678	724.01127	3.02613	0.29857
std	13.16145	2.82913	743.82868	1.96812	0.45768
min	18.00000	0.00000	0.00000	0.00000	0.00000
25%	36.00000	2.00000	148.75000	2.07775	0.00000
50%	51.00000	3.00000	376.00000	2.79200	0.00000
75%	57.00000	5.00000	1336.75000	3.75625	1.00000
max	63.00000	30.00000	2537.00000	18.43400	1.00000

Fig – 2

- The average age of a visitor is 46.2 years old.
- There is a standard deviation of 13.16 years.
- The minimum age is 18 years old, and the maximum age is 63 years old.
- 25% of visitors are younger than 36 years old and 75% of visitors are younger than 57 years old.
- The average visitor spends 724 seconds (about 12 minutes) on the website per visit.
- There is a standard deviation of 744 seconds.
- The minimum time spent on the website is 0 seconds and the maximum time spent is 2537 seconds (about 42 minutes).
- 25% of visitors spend 149 seconds (about 2.5 minutes) or less on the website and 75% of visitors spend 1337 seconds (about 22 minutes) or less on the website.
- The average visitor makes 3.57 website visits.
- There is a standard deviation of 2.83 visits.
- The minimum number of visits is 0 and the maximum number of visits is 30.

- 25% of visitors have made 2 or fewer visits and 75% of visitors have made 5 or fewer visits.
- The average visitor views 3.03 pages per visit.
- There is a standard deviation of 1.97 pages.
- The minimum number of page views is 0 and the maximum number of page views is 18.43.
- 25% of visitors view 2 or fewer pages per visit and 75% of visitors view 4 or fewer pages per visit.

Observations on age spectrum of visitors

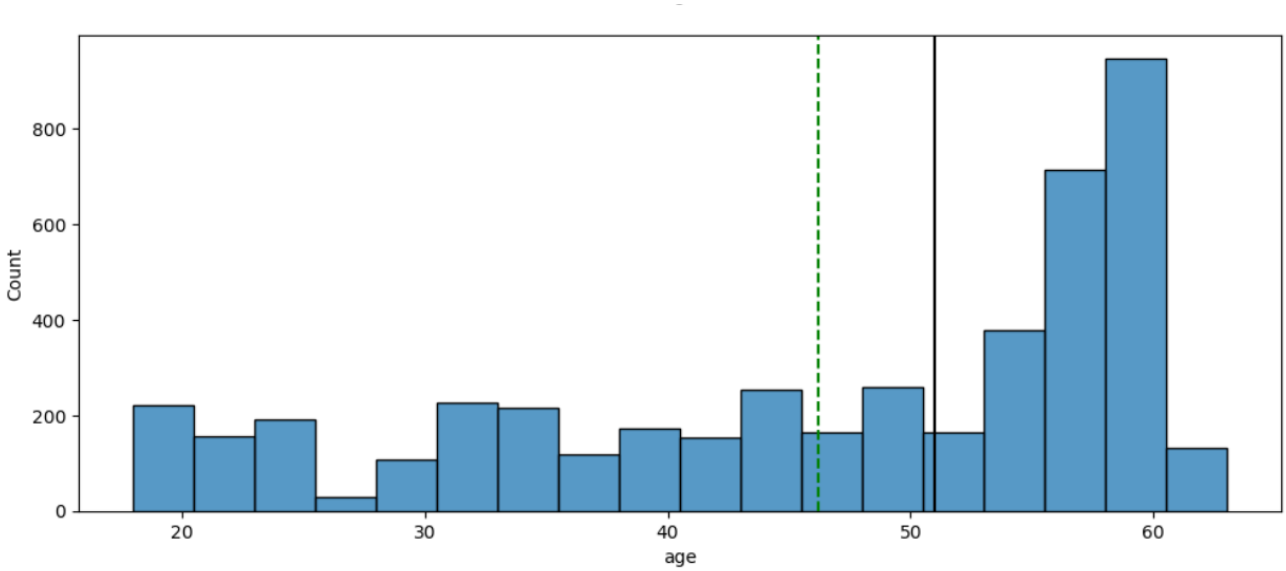


Fig – 3 Graph of age vs no. of leads

- The most common age group is 20-24 years old, followed by 25-29 years old and 30-34 years old.
- There is a small bump in the count around 45-49 years old, but the number of people then drops off sharply in the older age groups.
- The graph shows a right-skewed distribution, which means that there are more people in the younger age groups and the number of people decreases as age increases.
- There are relatively few people in the age groups below 20 and above 60.
- There is a steep decline in the number of people after the age of 60.

Observations related to current occupation.

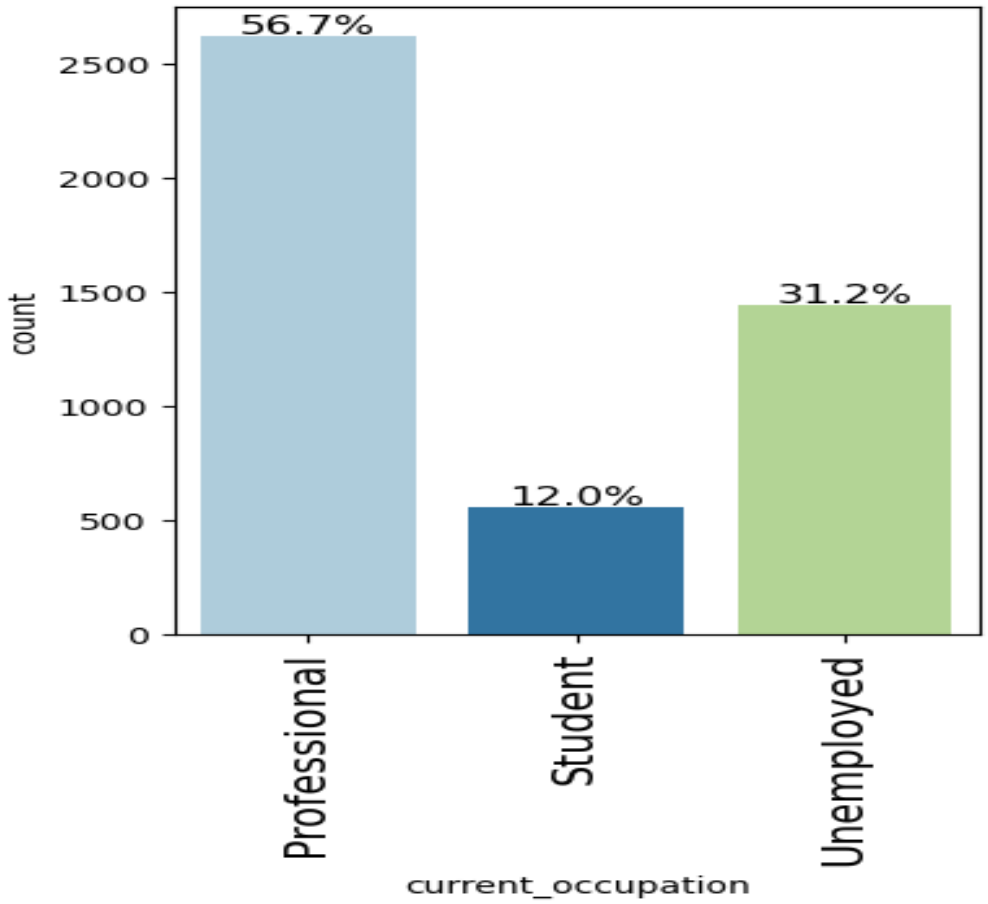


Fig – 4 graph of Current occupation Vs Leads

The data contains occupational details of leads with the company.

- 56.7% of people are currently employed as professionals.
- 31.2% of people are currently unemployed.
- 12.0% of people are currently students.

Professional 2616

Unemployed 1441

Student 555

Out of the 4612 leads 2616 are professional job holders, 1441 are unemployed and 555 are students.

First interactions

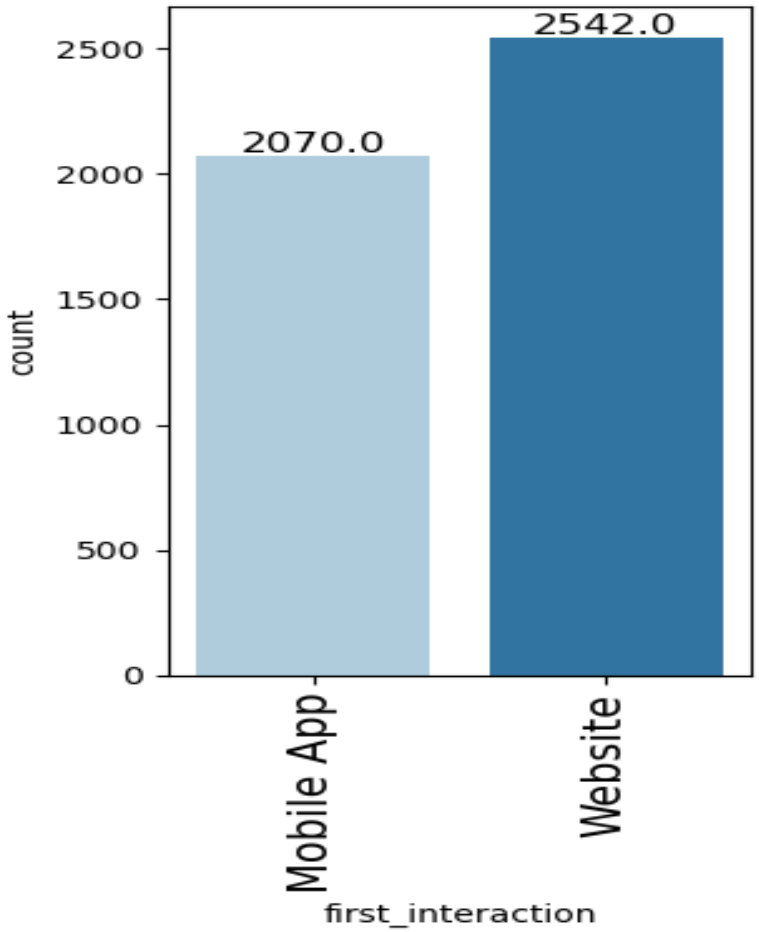


Fig – 5

- The most common type of first interaction is through WEBSITE with approximately 2542 interactions.
- The next most common type of first interaction is via MOBILE APP with approximately 2070 interactions.

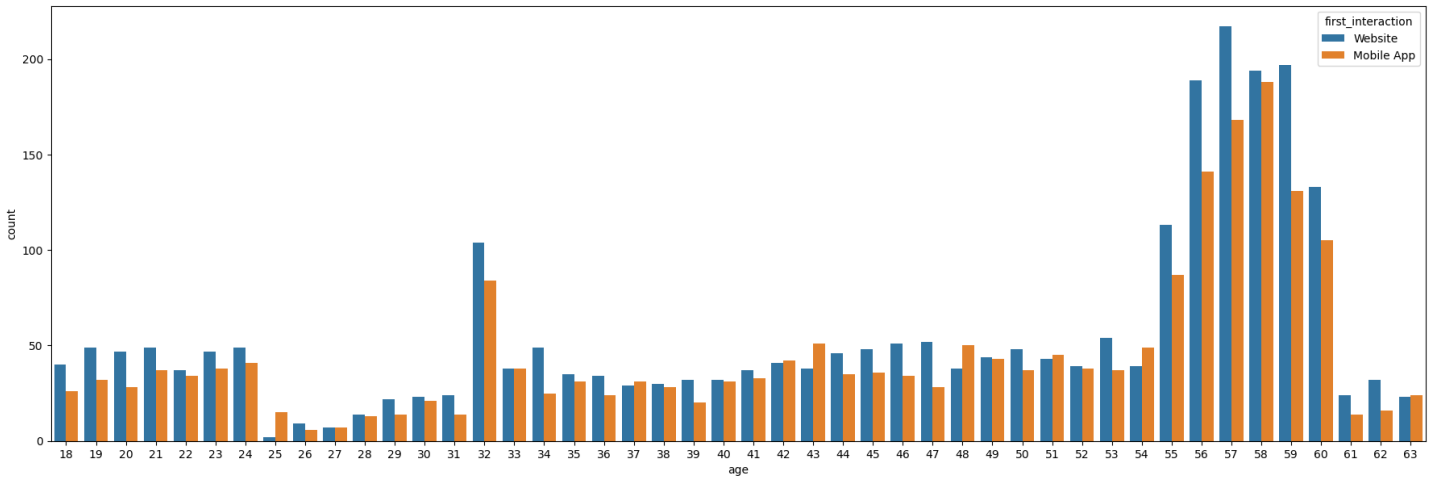


Fig – 6

- Website: There are more users who had their first interaction through the website than the mobile app, across all age groups.
- The distribution of users is pretty spread out in both modes with dips in app usage among ages of 20-35.
- There is a huge spike in users in both modes in the age group of 50-60

  Which means people are still trying to learn something new.

Observations on profile Completion

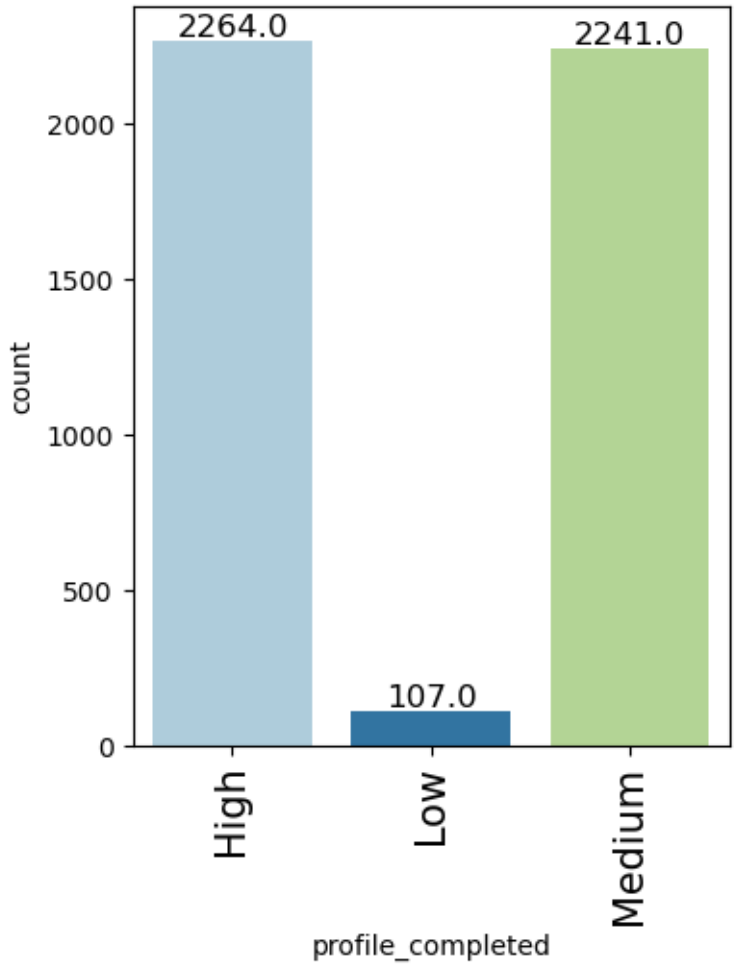


Fig – 7

Low - (), Medium - (50-75%), High (75-100%)

As you can see from the graph above

Out of the 4612 leads

2264 have completed their profiles 75-100%

While 0-50%

2241 have completed 50-75%

And 107 have finished about 0-50%

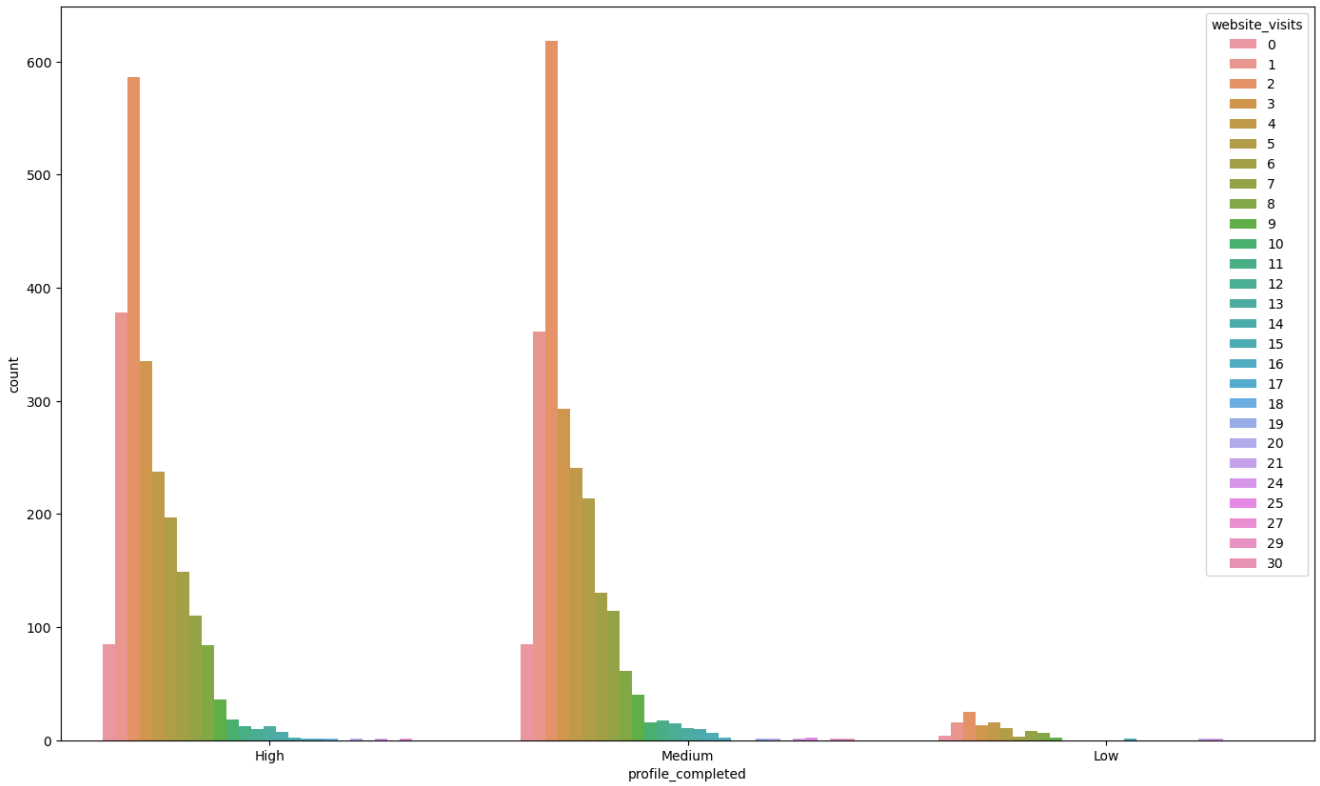


Fig – 8 Profile completion Vs website visits

- Website visits tend to be higher for users with higher profile completion. As the level of profile completion increases from "Low" to "High", the bars representing higher website visit counts become taller. This suggests that users who invest more effort in completing their profiles are also more likely to engage with the website by visiting it more frequently.
- There is some overlap in the distribution of website visits between the different profile completion groups. For example, there are some users in the "Low" profile completion group who have had 20 or more website visits, and there are some users in the "High" profile completion group who have had only 10 or fewer website visits. This suggests that profile completion is not the only factor that influences website visit count. Other factors, such as user interest and website features, may also play a role.
- Overall, this countplot suggests that there is a positive correlation between profile completion and website visit count. However, it is important to note

that correlation does not equal causation. Just because users with higher profile completion tend to have more website visits does not mean that completing one's profile necessarily leads to more website visits.

Observations on print_media

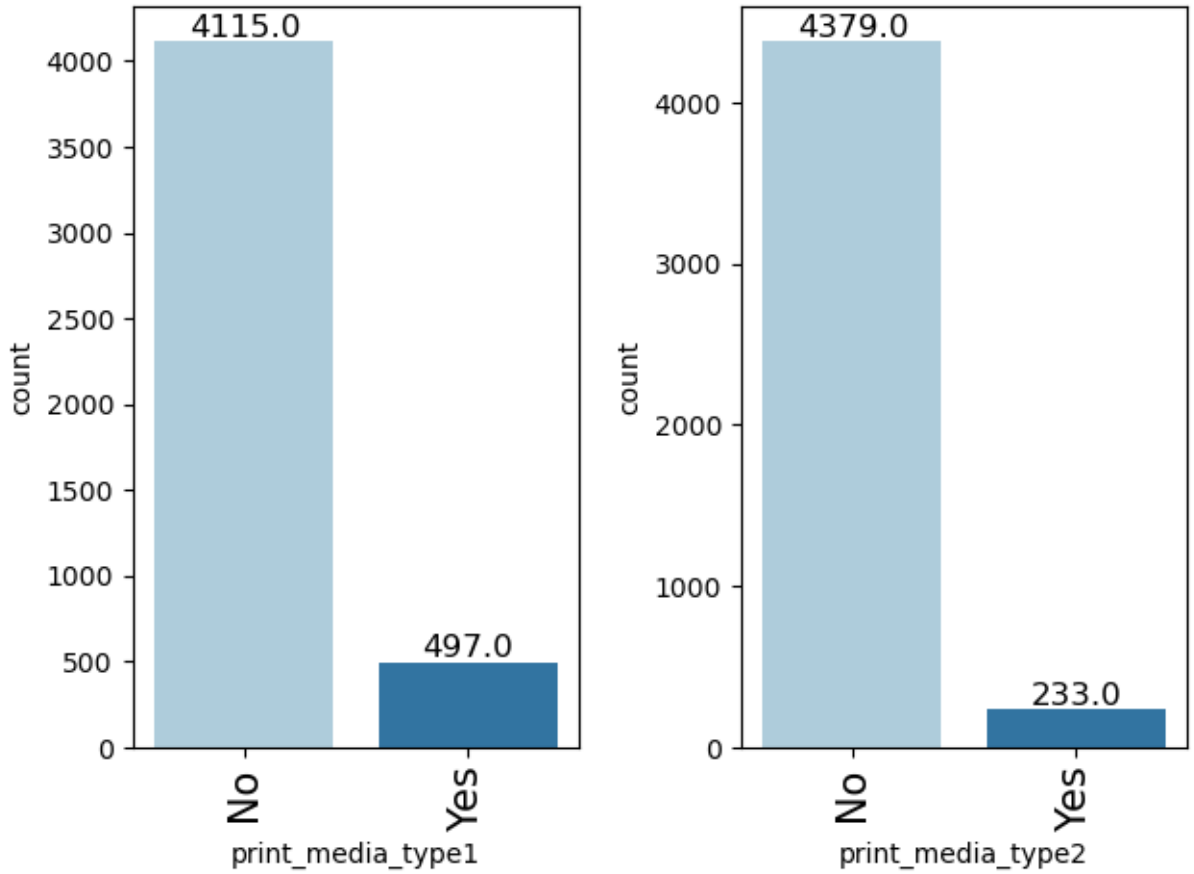


Fig – 9

From the above figures we can deduce the following

- Newspapers (print media 1) has not been a great help in marketing as it has proven to be not very effective as out of 4600 leads only 497 are positive about newspapers.

- Similar with Magazines (print media 2) most leads out of the group did not come from magazine ads.
- This suggests we might be targeting the wrong newspapers or magazines and may need to re-evaluate our options when we decide to post adverts again through print media.

Observations on Digital media

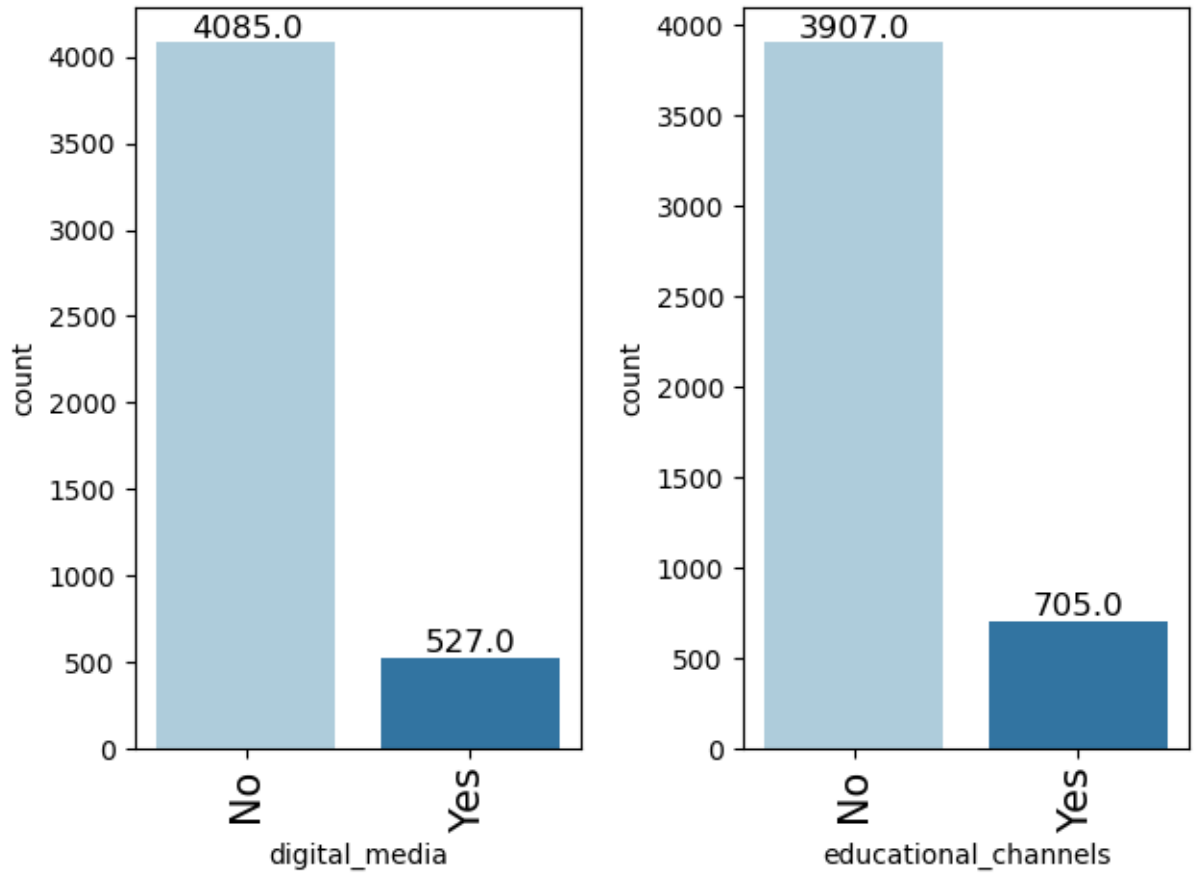


Fig – 10

From the figure above we can see the following

- The data shows both digital media and educational channels did not provide apt response.

- Most of the leads did not come through both these modes.
- Out of 4612 only 527 people were influenced by digital media.
- And only 705 were exposed to educational channels.
- This can definitely be improved, as the trend shows response to digital media and educational channels has been on the increase post covid.

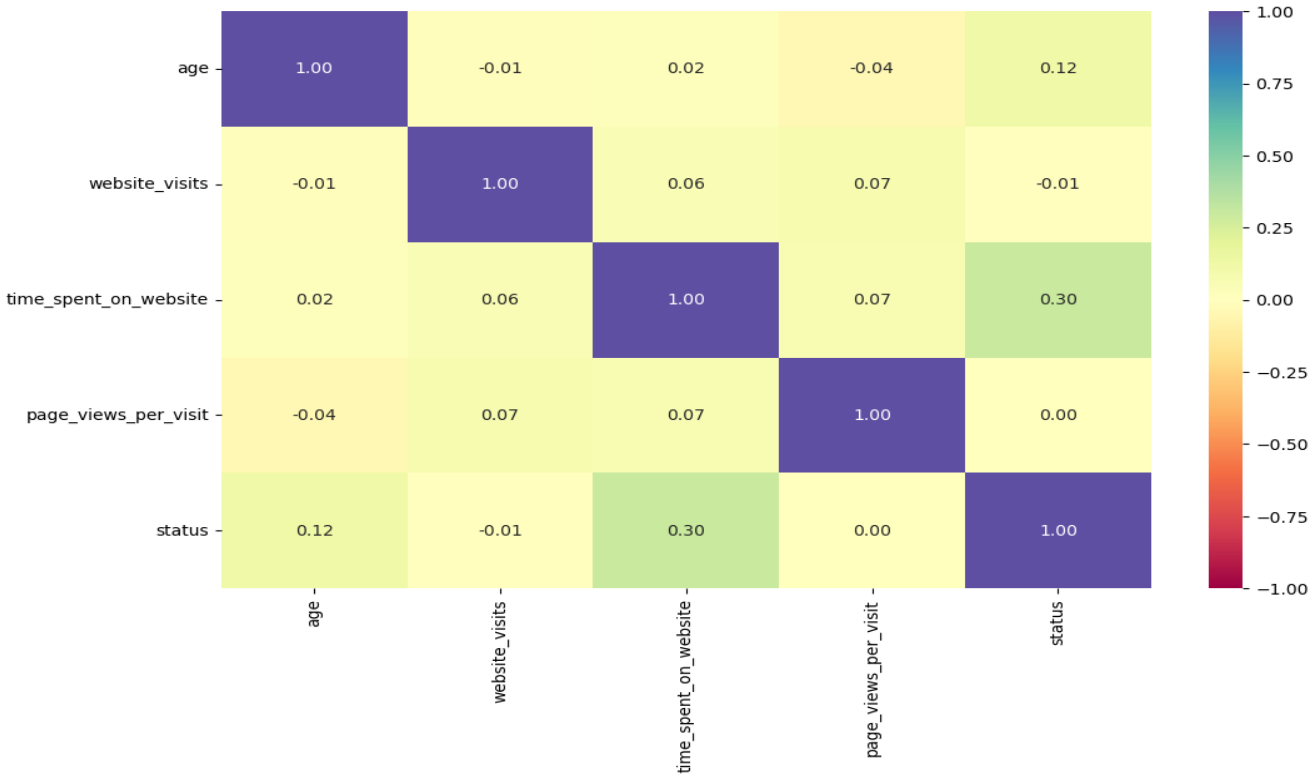


Fig – 11

- Profile completion: There seems to be a positive correlation between profile completion and conversion, as the "High" profile completion level shows reddish hues in the "Converted" column.
- Website visits and time spent: Both website visits and time spent on the website appear to have positive correlations with conversion.
- A weak correlation is observed between age and time spent on website.

- Age and website visits have a negative correlation.
- For older adults, websites might be a primary source of information and learning, as they might not be as familiar with or comfortable using alternative online platforms like social media or mobile apps.

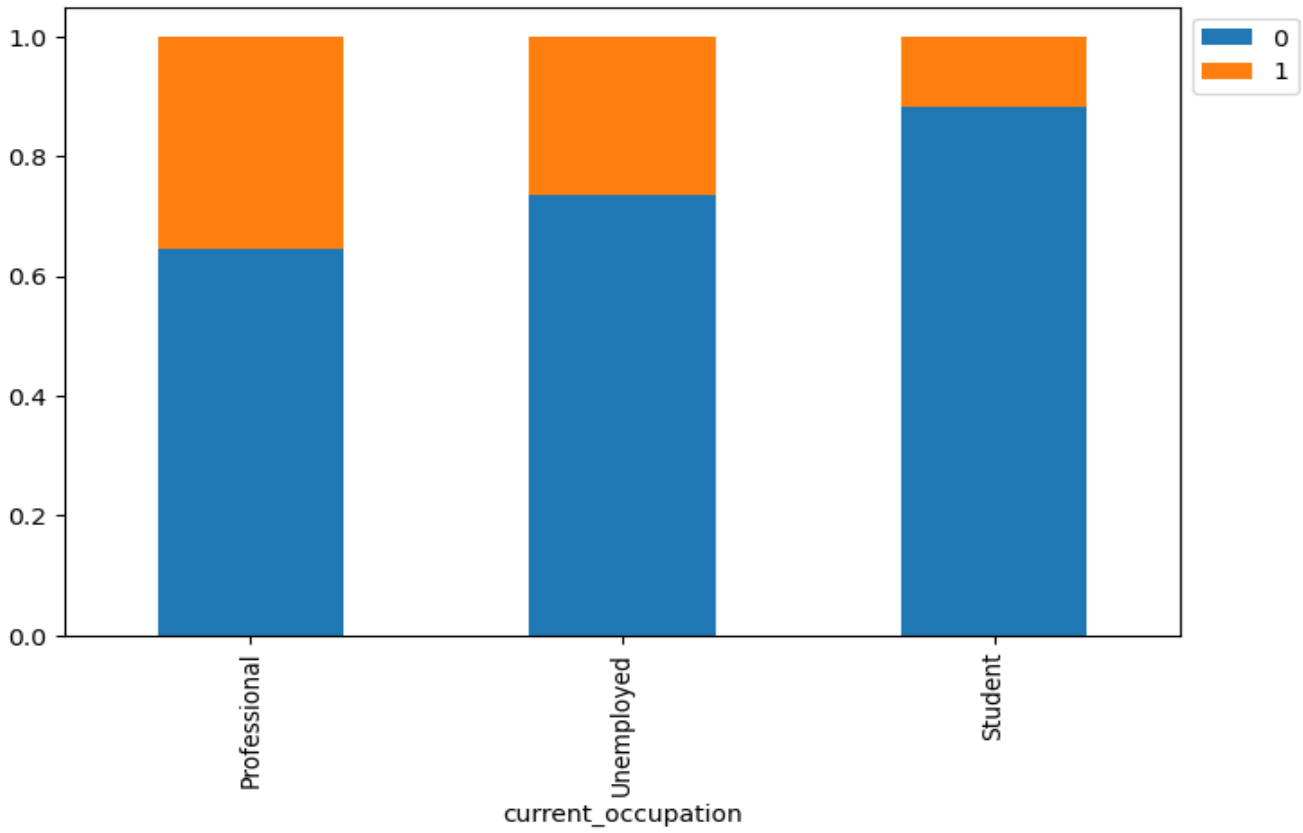


Fig – 12

- Overall conversion rate: Looking at the total number of leads for each status at the bottom (4612 for All, 1377 for Converted), we can see that the overall conversion rate is around 30% (1377 / 4612).
- Occupation and conversion:
    - Professionals: Have the highest overall number of leads (2616), but also the lowest conversion rate among the three groups (around 35%, 929 converted leads).

- Unemployed: Have the highest conversion rate (around 36%, 383 converted leads) despite having the second-lowest overall number of leads (1441).
- Students: Have the lowest overall number of leads (555) and a moderate conversion rate (around 12%, 65 converted leads).

Distribution of age across different occupational groups

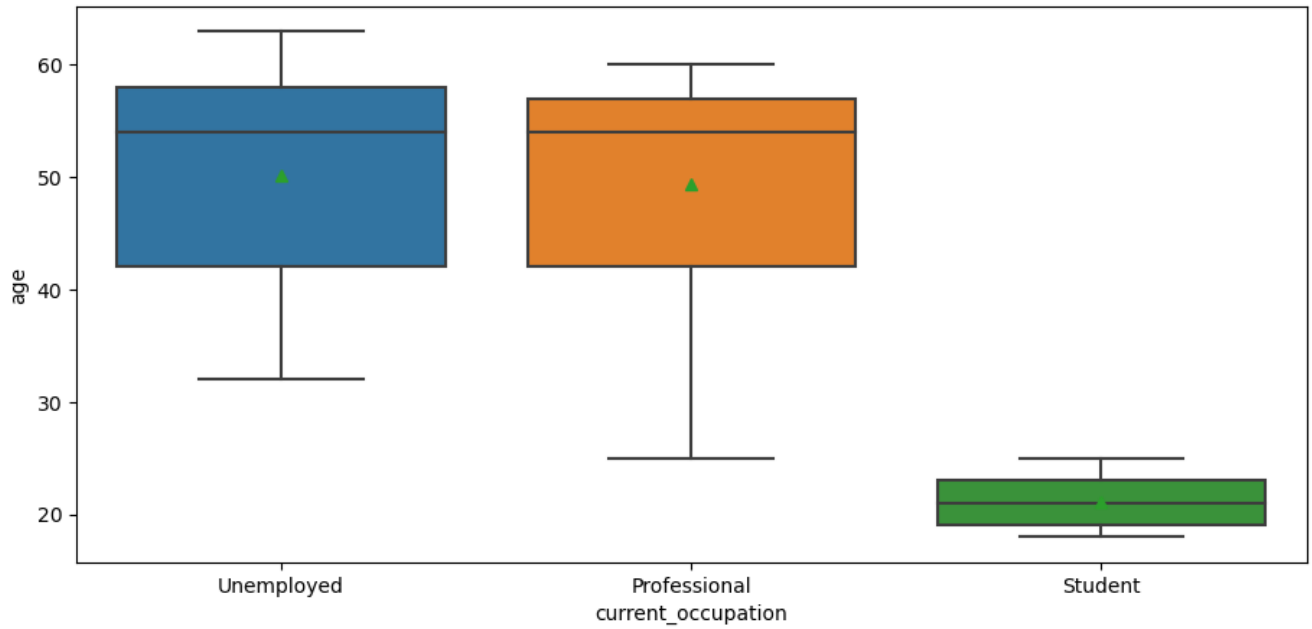


Fig – 13

- Comparison of medians and means: Interestingly, in all three groups, the mean age is slightly higher than the median age. This suggests that the distribution of ages within each group is slightly skewed towards older ages.
- Professionals:
  - Maintain the highest median and mean age among the three groups.
  - The small difference between their median and mean age indicates a symmetrical distribution of ages within this group.
- Unemployed:
  - Still have the lowest median and mean age.
  - The larger difference between their median and mean age compared to professionals suggests a slightly right-skewed distribution, with potentially more data points towards the younger end of the age range.

- Students:
    - Show a similar pattern to unemployed with a slightly right-skewed distribution based on the difference between median and mean.
    - Their median and mean age remain in between those of professionals and unemployed.

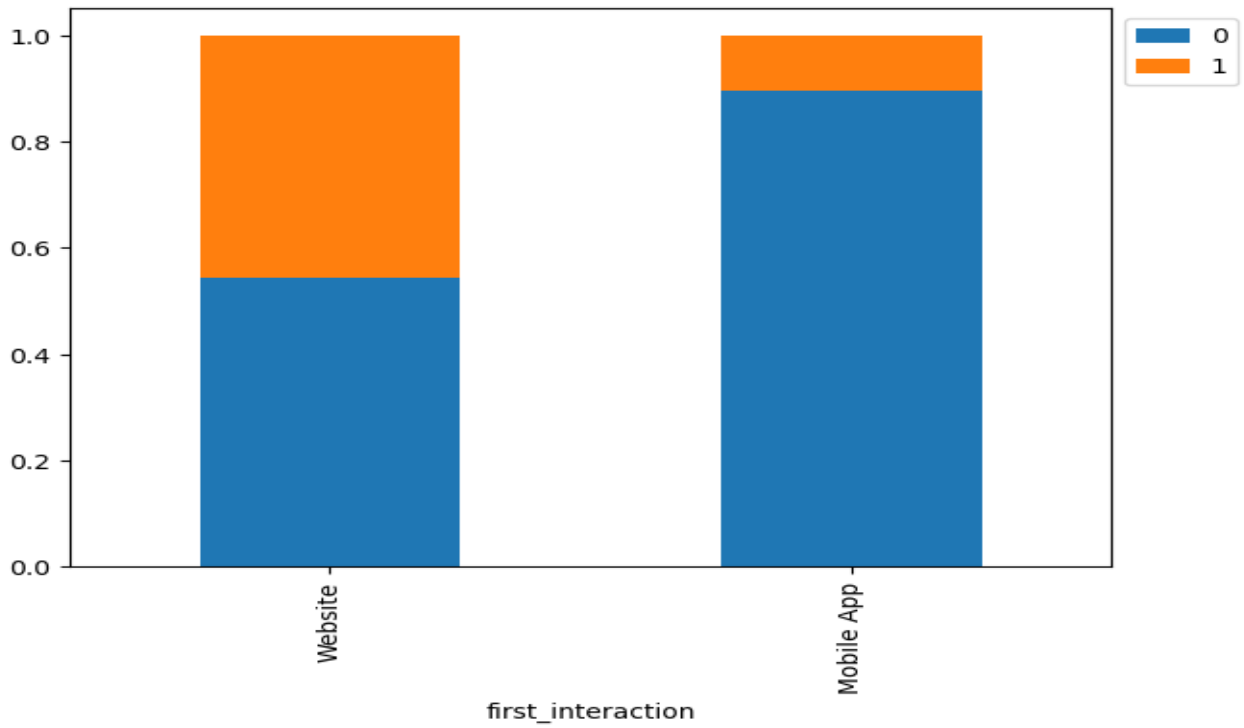


Fig – 14

- Overall conversion rate: Looking at the total number of leads for each status at the bottom (4612 for All, 1377 for Converted), we can see that the overall conversion rate is around 30% (1377 / 4612).
- First interaction and conversion:
    - Website: Has a higher total number of leads (2542) compared to the Mobile App (2070).
    - Website: Also has a higher overall conversion rate (around 46%, 1159 converted leads) compared to the Mobile App (around 11%, 218 converted leads).

- This suggests that leads who first interacted with ExtraaLearn through the website are more likely to convert to paid customers than those who first interacted through the Mobile App.

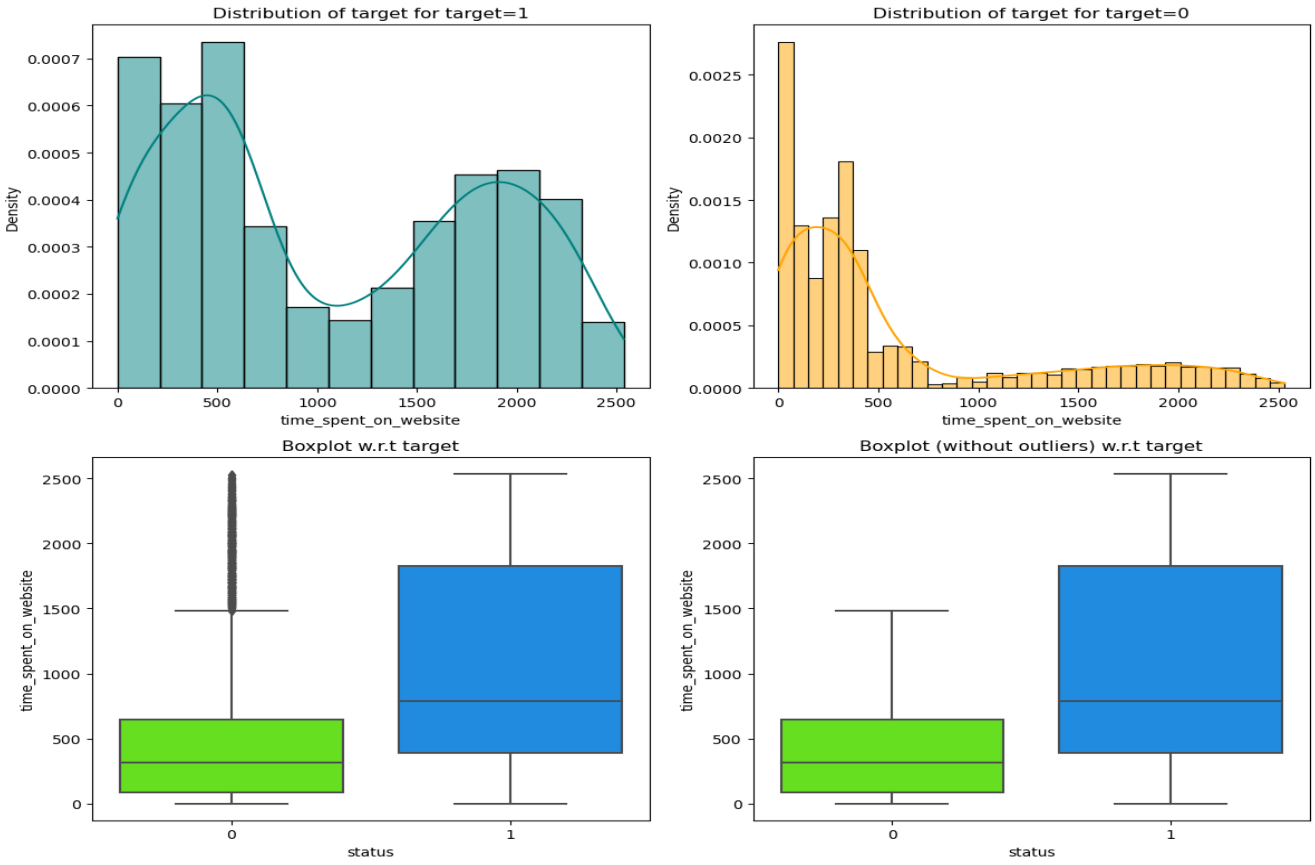


Fig – 15

Distribution of target for target=1 (Converted leads)

- The curve seems to be right-skewed, meaning there are more users who spent less time on the website and fewer who spent a longer time.
- The peak of the curve is around the 500-second mark, suggesting that most converted customers spent around 500 seconds (8 minutes) on the website.

- There appears to be a long tail extending towards the right, indicating that some converted customers spent significantly more time on the website.

Distribution of target for target=0 (non-converted leads)

- The curve also seems to be right-skewed, but with a steeper slope compared to the first graph. This suggests that a larger proportion of non-converted customers spent less time on the website.
- The peak of the curve is around the 250-second mark (4 minutes), indicating that most non-converted customers spent less time on the website than converted customers.
- The tail towards the right is shorter compared to the first graph, suggesting that fewer non-converted customers spent a very long time on the website.

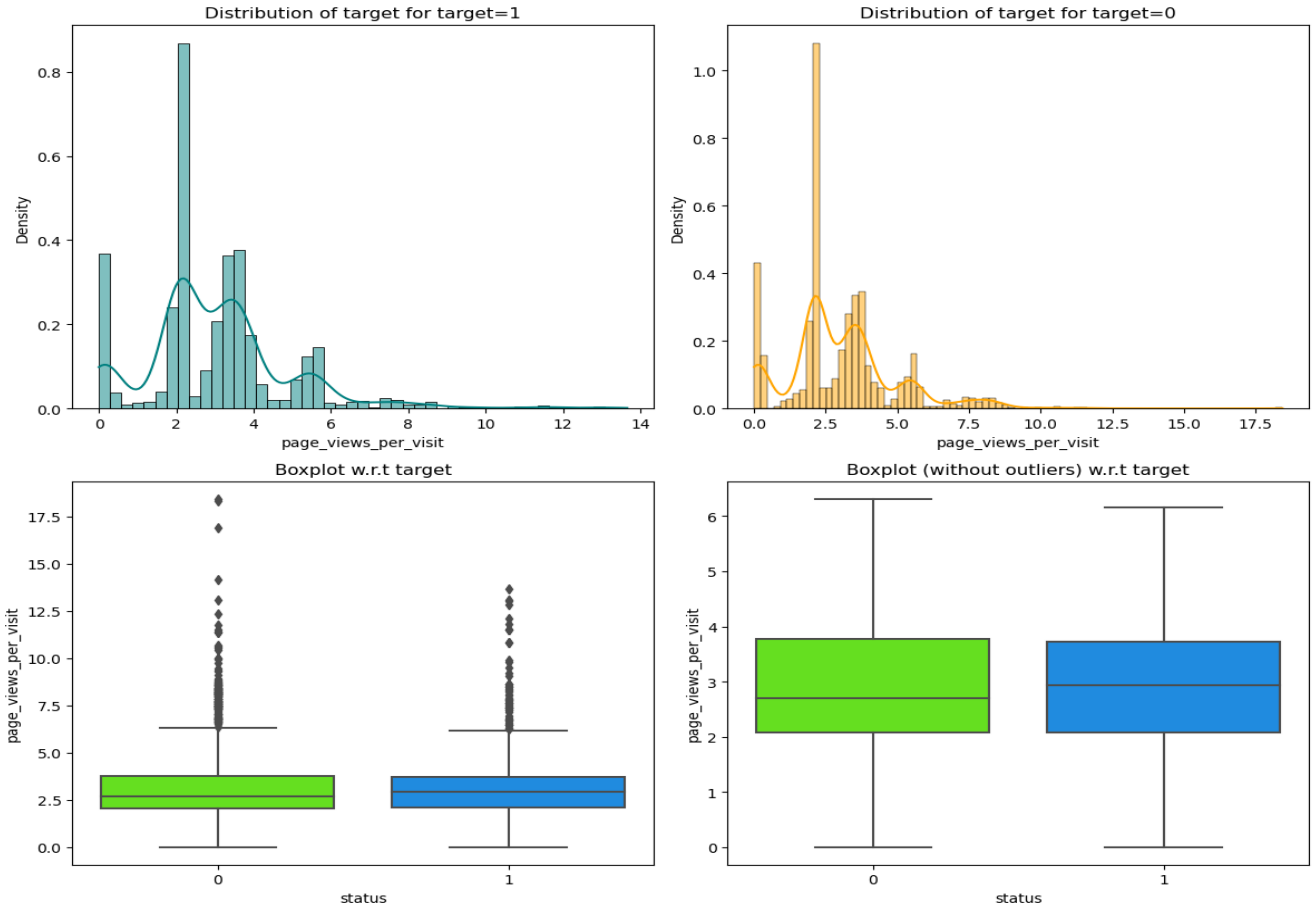


Fig – 16

- The curve seems to be right-skewed, meaning there are more users who viewed fewer pages per visit and fewer who viewed many pages.
- The peak of the curve is around 4-5 page views per visit, suggesting that most converted customers viewed around 4-5 pages during their visits.

- There appears to be a long tail extending towards the right, indicating that some converted customers viewed significantly more pages.
- The curve also seems to be right-skewed, but with a steeper slope compared to the first graph. This suggests that a larger proportion of non-converted customers viewed fewer pages per visit.
- The peak of the curve is around 2-3 page views per visit, indicating that most non-converted customers viewed fewer pages than converted customers.
- The tail towards the right is shorter compared to the first graph, suggesting that fewer non-converted customers viewed a very large number of pages.
- Converted customers (target=1) generally viewed more pages per visit than non-converted customers (target=0).
- The distribution of page views per visit is right skewed for both groups, but the non-converted customers have a steeper slope, indicating a more concentrated cluster of users who viewed very few pages.

Observations on profile completion with respect to conversion

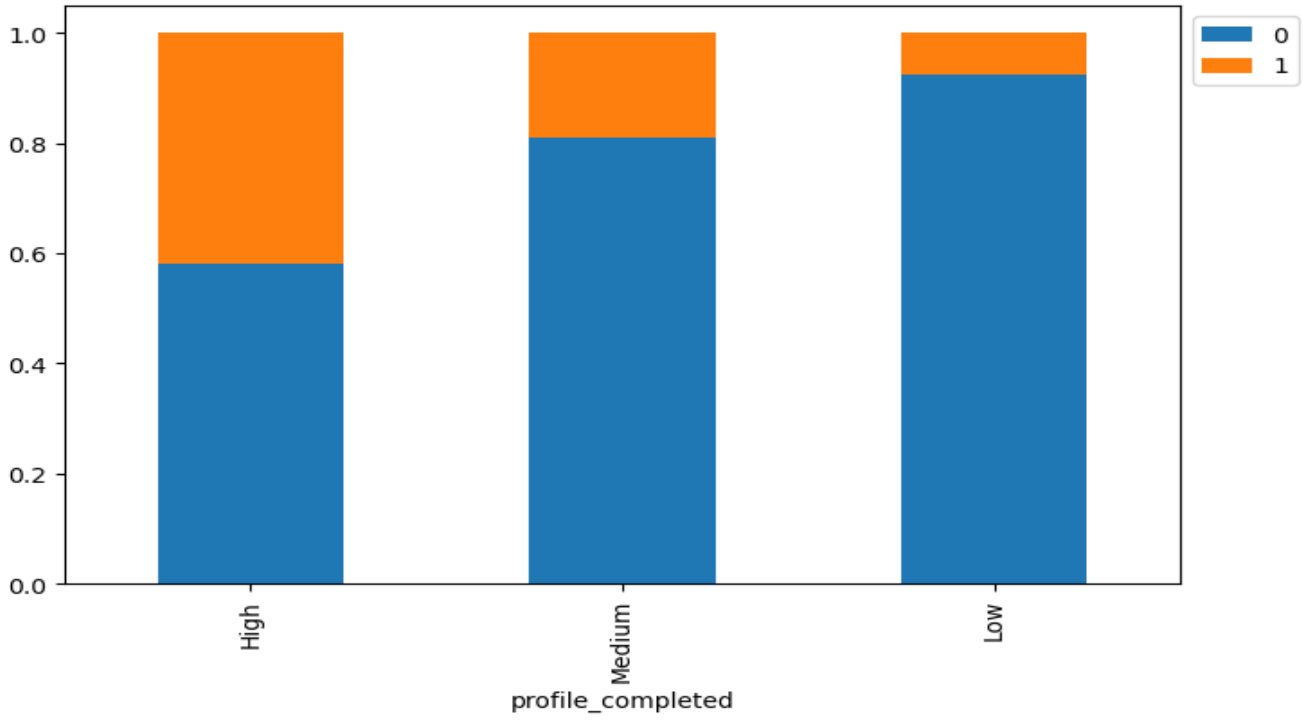


Fig – 17

- High profile completion:
  - Has the highest overall number of leads (2264).

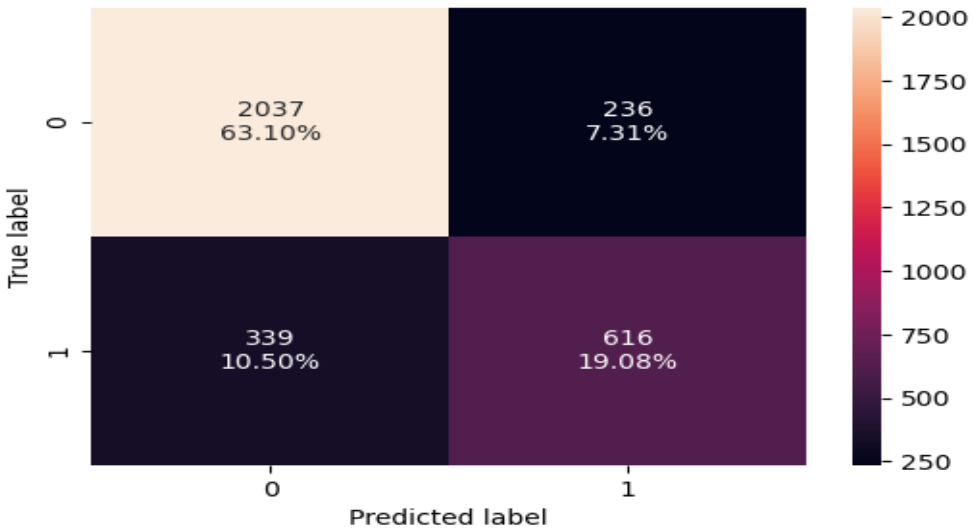
  - Also has the highest conversion rate, with around 72% (946 converted leads) of leads who completed a high percentage of their profiles becoming paid customers.

- Medium profile completion:
  - Has the second-highest overall number of leads (2241).
  - Has a moderate conversion rate, with around 23% (423 converted leads) of leads who completed a medium percentage of their profiles becoming paid customers.
- Low profile completion:
  - Has the lowest overall number of leads (107).
  - Also has the lowest conversion rate, with only around 8% (8 converted leads) of leads who completed a low percentage of their profiles becoming paid customers.

Overall observation:

- Leads who completed a higher percentage of their profiles are more likely to convert to paid customers.

Checking model performance on training set



	Accuracy	Recall	Precision	F1
0	0.82187	0.64503	0.72300	0.68179

Fig – 18

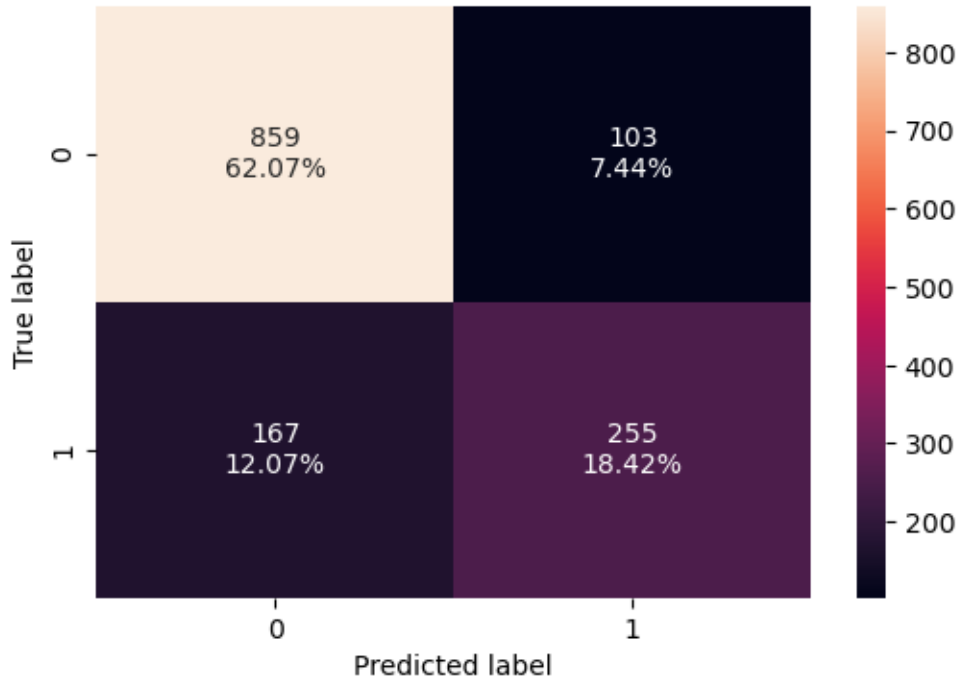
- Accuracy: 0.8218 - This represents the overall percentage of correctly classified instances in the training set. It's a good general indicator of performance but can be misleading in imbalanced datasets (like yours) where the majority class is easier to predict.
- Recall (for class 1): 0.6450 - This measures the model's ability to correctly identify instances that belong to the minority class (class 1). It's important to consider this metric alongside accuracy, especially when dealing with imbalanced data.
- Precision (for class 1): 0.7230 - This measures the proportion of instances predicted as class 1 that are true class 1. A high precision indicates the model is not making many false positives for the minority class.
- F1-score (for class 1): 0.6817 - This is a harmonic mean of recall and precision, providing a balanced measure of the model's performance on both aspects.

Interpretation:

- Overall, the accuracy of 82% seems good, but it's important to consider the class imbalance. Since the majority class (class 0) likely dominates the data, the model could simply be predicting the majority class for most instances and still achieve a high accuracy.
- The recall of 64% for class 1 is moderate. This means the model identifies a bit over half of the actual class 1 instances correctly. It's good to see that the model is not completely neglecting the minority class.

- The precision of 72% for class 1 is relatively high. This indicates that most of the instances predicted as class 1 are true class 1. This is encouraging, as it suggests the model is not making many false positives for the minority class.
- The F1-score of 68% is a decent balance between recall and precision. It further confirms that the model is performing reasonably well on both aspects for the minority class.

Checking model performance on test set



	Accuracy	Recall	Precision	F1
0	0.80491	0.60427	0.71229	0.65385

Fig – 19

- Accuracy: 0.8049 - This represents the overall percentage of correctly classified instances on the test set. It's slightly lower than the accuracy on the training set (0.8218), which could indicate some overfitting on the training data.
- Recall (for class 1): 0.6042 - This is moderate, similar to the recall on the training set (0.6450). It means the model identifies around 60% of the actual class 1 instances correctly.
- Precision (for class 1): 0.7122 - This is slightly higher than the precision on the training set (0.7230). It suggests the model makes fewer false positives for the minority class (class 1) on the test set.
- F1-score (for class 1): 0.6538 - This is a balanced measure of recall and precision, and it's slightly lower than the F1-score on the training set (0.6817).

Interpretation:

- Compared to the training set, the model's performance on the test set is slightly lower in accuracy and F1-score, but similar in recall and slightly higher in precision for class 1. This suggests some potential overfitting on the training data but overall decent generalizability.
- The moderate recall for class 1 indicates the model still misses some true minority class instances, which could be an area for improvement.
- The relatively high precision for class 1 is encouraging, as it shows the model makes fewer false positives for the minority class on the test set.

Recommendations:

- Consider techniques to reduce overfitting, such as regularization or early stopping.
- Explore alternative algorithms less prone to overfitting, such as Support Vector Machines with appropriate cost functions.
- Investigate why the model misses some true class 1 instances and try to improve its recall for the minority class.

ROC-AUC

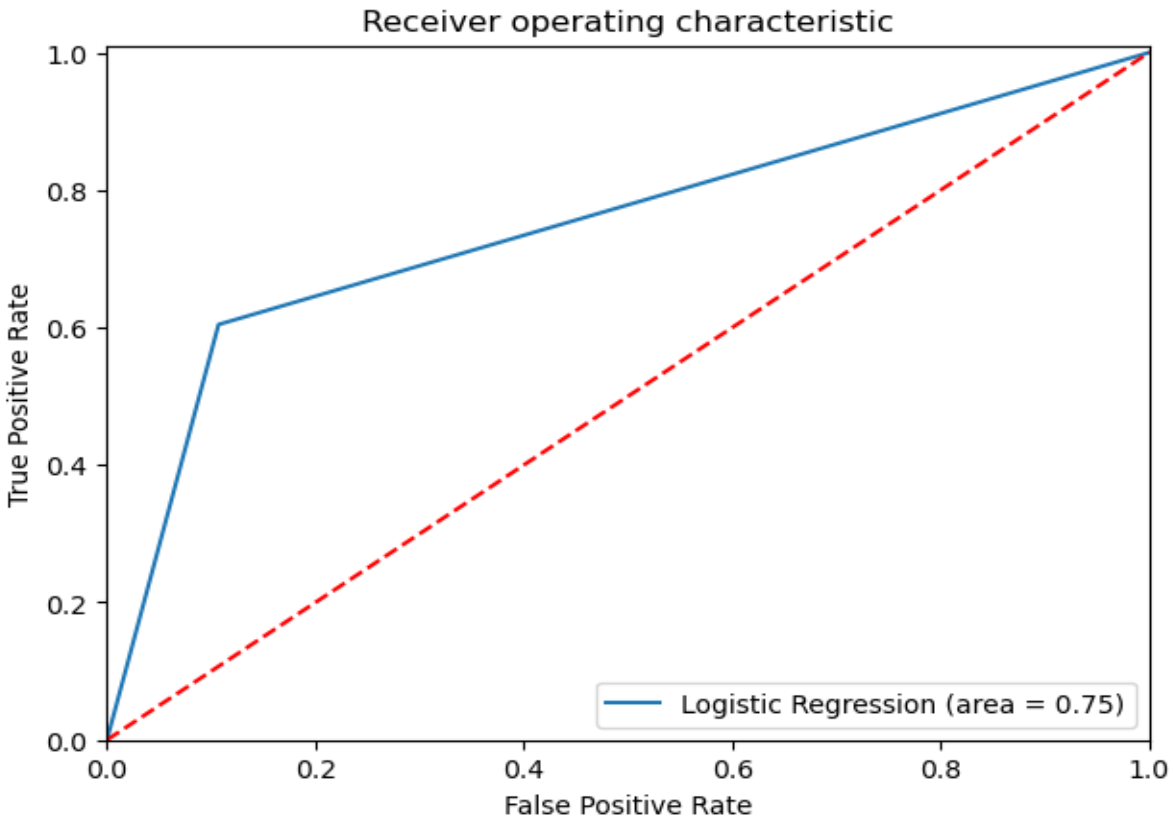


Fig – 20

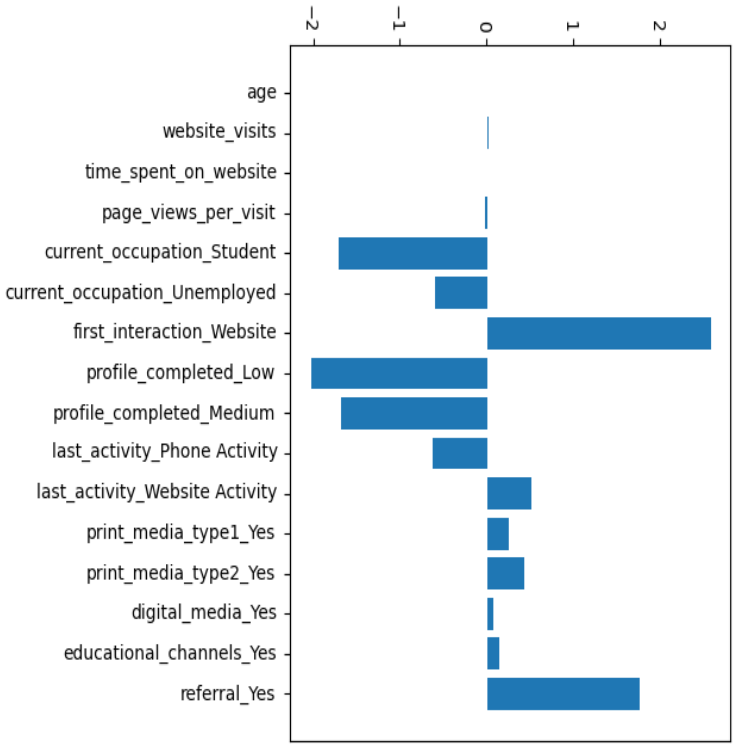
Train ROC-AUC score is : 0.8738226805453503

Test ROC-AUC score is : 0.8562249362012394

- Both the train and test ROC-AUC scores are good, indicating that your model can effectively distinguish between the positive and negative classes.
- However, the train ROC-AUC score is slightly higher than the test ROC-AUC score. This suggests a potential for overfitting, where the model memorizes the training data too well and doesn't generalize well to unseen data.

Inferences post ROC - AUC

- ROC-AUC scores:
    - Area Under the Receiver Operating Characteristic Curve (AUC-ROC) is a robust metric for binary classification, independent of class imbalance.
    - Scores above 0.8 are considered good, especially for imbalanced datasets.
    - Your model's scores suggest good ability to distinguish between classes on both training and test sets.
- Metrics on Training and Test sets:
    - Training set metrics generally higher than test set, confirming potential overfitting.
    - Training set:
        - Accuracy: 0.8218 (overall correctly classified instances)
        - Recall for class 1: 0.6450 (correctly identified minority class instances)
        - Precision for class 1: 0.7230 (true positives to predicted positives for class 1)
        - F1-score for class 1: 0.6817 (balanced measure of recall and precision)
    - Test set:
        - Accuracy: 0.8049 (slightly lower than training)
        - Recall for class 1: 0.6042 (lower than training, indicating improved focus is needed)
        - Precision for class 1: 0.7122 (slightly higher than training)
        - F1-score for class 1: 0.6538 (lower than training)



Feature: age, Score: 0.00112
Feature: website_visits, Score: 0.00633
Feature: time_spent_on_website, Score: 0.00130
Feature: page_views_per_visit, Score: -0.02753
Feature: current_occupation_Student, Score: -1.71345
Feature: current_occupation_Unemployed, Score: -0.60896
Feature: first_interaction_Website, Score: 2.57413
Feature: profile_completed_Low, Score: -2.03226
Feature: profile_completed_Medium, Score: -1.68960
Feature: last_activity_Phone Activity, Score: -0.63192
Feature: last_activity_Website Activity, Score: 0.49707
Feature: print_media_type1_Yes, Score: 0.24218
Feature: print_media_type2_Yes, Score: 0.42313
Feature: digital_media_Yes, Score: 0.05850
Feature: educational_channels_Yes, Score: 0.13176
Feature: referral_Yes, Score: 1.75371

- Negative scores: Most features have negative scores, indicating they are negatively correlated with the target variable (status). This means that higher values of these features are associated with a lower probability of conversion.
- Positive scores: A few features have positive scores, suggesting they are positively correlated with the target variable. Higher values of these features are associated with a higher probability of conversion.

- Top features:
    - first_interaction_Website has the highest positive score (2.60241), indicating that users who first interacted with the website are more likely to convert.
    - current_occupation_Student has the most negative score (-2.09371), suggesting that students are less likely to convert compared to other occupations.
    - page_views_per_visit has a significant negative score (-0.03821), implying that users who view fewer pages per visit are more likely to convert.

Model Comparisions:

Training performance comparison:

	Logistic Regression	Logistic Regression After Tuning	LDA	LDA After Tuning	Decision Tree	Decision Tree After Tuning
Accuracy	0.82187	0.82776	0.82404	0.82404	1.00000	0.83798
Recall	0.64503	0.65969	0.65969	0.65969	1.00000	0.89005
Precision	0.72300	0.73171	0.72165	0.72165	1.00000	0.67035
F1	0.68179	0.69383	0.68928	0.68928	1.00000	0.76473

Fig – 22

Models tested:

- Logistic Regression: Both before and after tuning (hyperparameter optimization).
- LDA: Linear Discriminant Analysis, another linear model suitable for two-class problems.
- Decision Tree: A non-linear model capable of capturing complex relationships in the data.

Metrics:

- Accuracy: Overall percentage of correctly classified instances.
- Recall: Proportion of true class 1 instances correctly identified.
- Precision: Proportion of predicted class 1 instances that are actually true class 1.
- F1-score: Harmonic mean of recall and precision, balancing both aspects.

Observations:

- All models achieve high accuracy, above 82%: This suggests the data might be relatively easy to classify and all models capture some useful information.
- Tuning slightly improves Logistic Regression and Decision Tree: This signifies that fine-tuning hyperparameters can provide a small performance boost for these models.

- LDA and tuned Decision Tree achieve slightly higher recall: These models identify more true class 1 instances correctly compared to the other models.
- Logistic Regression after tuning has the highest precision: This means it makes fewer false positives for the minority class (class 1).
- Tuned Decision Tree has the highest F1-score: This indicates a good balance between recall and precision for this model.

Overall:

- All models perform well, offering different strengths and weaknesses.
- Logistic Regression offers good overall accuracy and precision, particularly after tuning.
- LDA and tuned Decision Tree are slightly better at identifying true class 1 instances but sacrifice some precision.
- Tuned Decision Tree achieves the best balance between recall and precision, as indicated by the F1-score.

Test performance comparison:

```
Test performance comparison:
           Logistic Regression  Logistic Regression After Tuning     LDA  \
Accuracy               0.80491                           0.82776 0.82404
Recall                 0.60427                           0.65969 0.65969
Precision              0.71229                           0.73171 0.72165
F1                     0.65385                           0.69383 0.68928

           LDA After Tuning  Decision Tree  Decision Tree After Tuning
Accuracy            0.82404        1.00000                     0.83598
Recall              0.65969        1.00000                     0.86730
Precision           0.72165        1.00000                     0.68156
F1                  0.68928        1.00000                     0.76330
```

Fig – 23

- Logistic Regression (LR): Baseline model without hyperparameter tuning.

- Logistic Regression After Tuning (LR T): LR with optimized hyperparameters.
- LDA: Linear Discriminant Analysis, another linear model suitable for two-class problems.
- LDA After Tuning (LDA T): LDA with optimized hyperparameters.
- Decision Tree (DT): Non-linear model capable of capturing complex relationships.
- Decision Tree After Tuning (DT T): DT with optimized hyperparameters.

Observations:

- Accuracy:
  - All models score above 80%, suggesting good overall classification ability.
  - Tuning slightly improves LR and LDA, but Decision Tree achieves a perfect score (1.00) after tuning, which might be overfitting on the test set.
- Recall:
  - DT and DT T excel in identifying true class 1 instances (higher Recall), but at the expense of precision.
  - LDA and LDA T also have decent Recall, while LR and LR T are the lowest.
- Precision:
  - LR and LR T show the highest Precision, meaning they make fewer false positives for the minority class (class 1).
  - LDA and LDA T are slightly lower, followed by DT and DT T, which sacrifice Precision for higher Recall.
- F1-score:
  - DT T achieves the highest F1-score, indicating a good balance between Recall and Precision.

- LDA and LDA T are next, followed by LR T and LR.

Feature Importance:

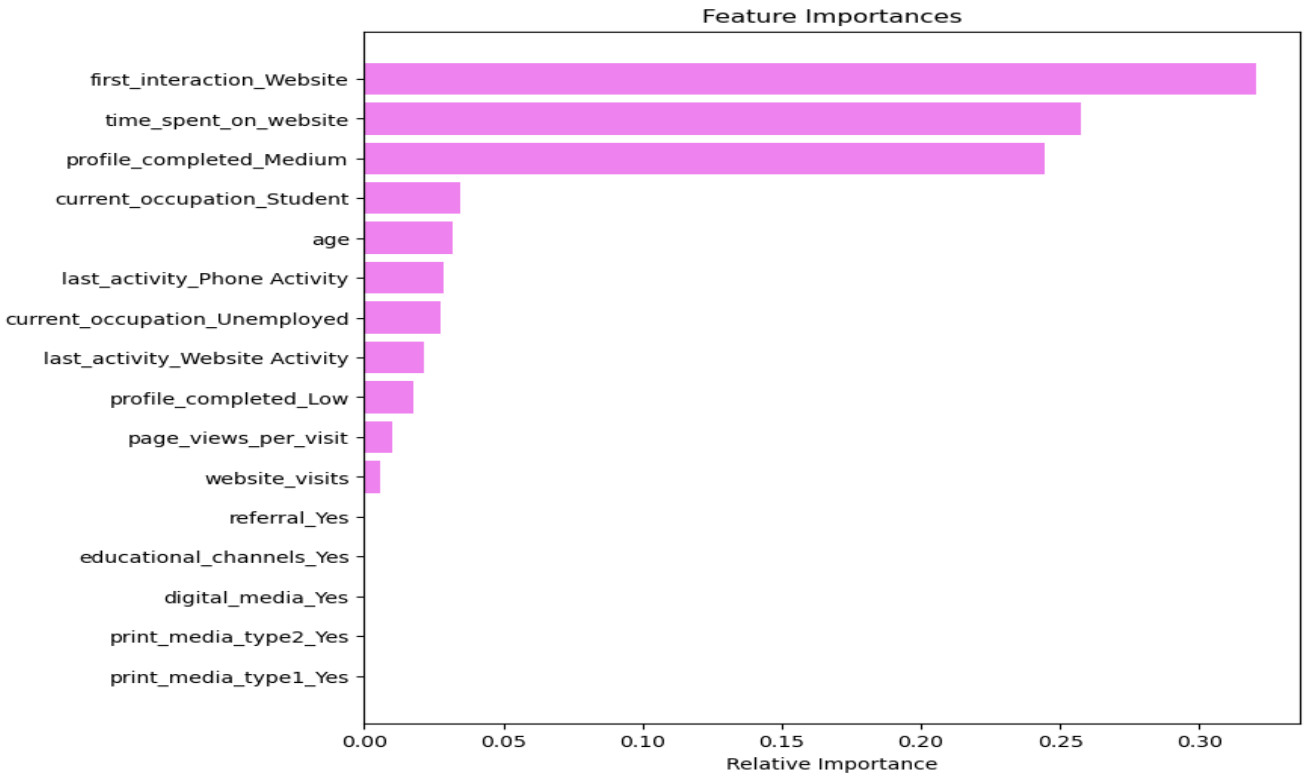


Fig – 24

- The top features, first_interaction_Website, Profile_completed_medium, age, and Time_spent_on_website, seem to have the most significant influence on the model's predictions.

- Features like current_occupation_Unemployed, referral_Yes, and Last _activity_website also appear to be moderately important.

- Features like time_spent_on_website, page_views_per_visit, and profile_completed_Medium, educational_channels_Yes have lower importance scores.

Decision Tree:

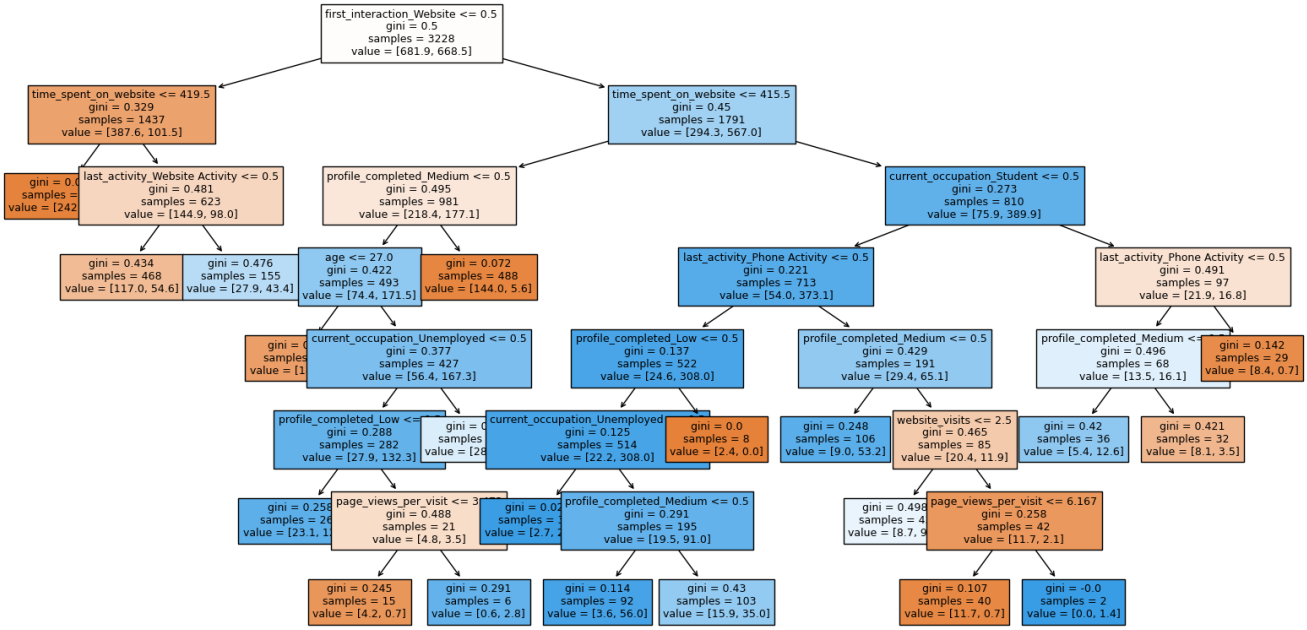


Fig – 25

first_interaction_Website <= 0.50: If the user's first interaction was not through the website, a further split based on time_spent_on_website follows. Higher time spent leads to class 0 (potentially low conversion) unless the last activity was on the website, then class 1 (potential conversion) becomes more likely.

first_interaction_Website > 0.50: If the user first interacted through the website, more complex branching occurs based on a combination of features like profile_completed_Medium,age, current_occupation_Unemployed, profile_completed_Low, page_views_per_visit, current_occupation_Student, and last_activity_Phone Activity. This suggests these features interact in determining conversion probability.

Business Recommendations

- Focus on user acquisition via the website: The initial split highlights the importance of website engagement for potential conversion. Consider optimizing website landing pages and user experience to encourage initial interaction.
- Analyze user segments based on branching points: Each branch represents a specific user segment with distinct conversion probabilities. Identify high-converting segments based on feature combinations and tailor marketing or outreach strategies accordingly.
- Investigate features with high decision power: Features like age, current_occupation, and profile completeness appear influential in later splits. Analyze these features' relationships with your target variable and optimize user profiling or targeting based on the insights gained.
- Consider model interpretability limitations: While decision trees offer some interpretability, their complexity can limit deep understanding. Explore other models like logistic regression for potentially clearer feature importance insights.